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Profile

Data Science and AI engineer with hands-on experience in NLP, deep learning, RAG pipelines, and end-to-end ML systems, currently completing a Master's in Advanced Data Science and AI at the University of Liverpool. Backed by 3+ years as a full-stack developer in the healthcare and pharmaceutical domain, bringing strong software engineering practices to ML pipeline development and deployment

Education

Sep 2025 – Sep 2026
Liverpool, United Kingdom

Master of Advanced Data Science and Artificial Intelligence

University of Liverpool [🌐](#)

- Engineered disease-spreading simulation in C and implemented parallelization and optimization to achieve 28x speedup with OpenMP and 31x speedup with MPI
- Collaborated on a group project to build a full-stack financial crisis analysis tool, analysing 3,158 US equity tickers across crisis periods using Next.js, FastAPI, and LLM-generated insights
- Designed and analyzed algorithms for sorting and optimization problems, including pancake-sorting strategies and Hamming-code based encoding schemes

Jul 2018 – May 2022
Chennai, India

Bachelor of Technology in Computer Science and Engineering

B.S. Abdur Rahman Crescent Institute of Science and Technology [🌐](#)

- First Class with Distinction
- CGPA 8.73/10

Professional Experience

Jun 2022 – Aug 2025
Chennai, India

Associate Engineer

Virtusa Consulting Services [🌐](#)

- Executed the front-end migration of 3 healthcare applications from a third-party library to a custom in-house library, delivering across a 12-month timeline
- Designed and developed reusable UI components in Storybook using React, Tailwind CSS, and Recoil to fill coverage gaps identified during migration, ensuring full component parity across the application suite
- Resolved 30+ bugs during the post-migration stabilization phase in close coordination with the QA team, earning a Star Performer of the Month award
- Advanced from junior developer in a two-person team to sole developer on a dedicated project, owning requirements gathering, feasibility analysis, feature demos, and sprint delivery
- Conducted technical interviews for prospective hires and mentored new joiners in React to ensure readiness for client-facing project assignments

Projects

Healthcare Premium [🌐](#)

Regression model that predicts annual health insurance premiums from user inputs

- Developed an end-to-end ML pipeline with XGBoost to predict healthcare insurance premiums, achieving $R^2 \approx 0.998$ on unseen data
- Applied age-stratified modelling with separate models for young and older cohorts after error analysis revealed distinct pricing patterns

- Constructed a composite medical risk score from 4 chronic condition indicators to improve accuracy
- Launched a Streamlit app that routes user inputs to age-stratified models and returns premium estimates
- github.com/ahmadfahim2k/ml-project-healthcare-premium 

Credit Risk Modelling

ML-powered web app that predicts loan default risk and generates a credit score with rating

- Built a credit risk prediction model using Logistic Regression with SMOTE and Optuna tuning, achieving ~93% accuracy on 50,000 loan records
- Engineered features loan-to-income and delinquency ratios, from merged customer, loan, credit bureau datasets
- Applied VIF and Information Value analysis to reduce multi-collinearity and select the most predictive features
- Converted default probability into a 300–900 credit score with a 4-tier rating (Poor/Average/Good/Excellent)
- Deployed an interactive Streamlit web app for real-time credit risk assessment
- github.com/ahmadfahim2k/ml-project-credit-risk-model 

Vehicle Damage Prediction

Deep learning system that classifies vehicle damage from images into 6 categories

- Trained a vehicle damage image classifier using transfer learning with ResNet50 in PyTorch, achieving ~81% validation accuracy across 6 damage categories
- Tuned hyperparameters with Optuna and applied data augmentation to improve model generalization
- Served the model via a Streamlit app and FastAPI REST API for real-time inference on uploaded vehicle images
- github.com/ahmadfahim2k/dl-project-vehicle-damage-detection 

ShoeBot : E-Commerce Chatbot

LLM-based shoe shopping assistant that handles product search, FAQs through a conversational interface

- Implemented semantic router using HuggingFace to classify user intent and dispatch queries to focused models
- Created a natural language to SQL pipeline using Groq to query a web-scraped Flipkart product catalog
- Integrated FAQ retrieval using RAG with ChromaDB and deployed the chatbot as a Streamlit application
- github.com/ahmadfahim2k/shoebot-genai 

Stock Analyzer : Financial Crisis Analysis Tool

Full-stack web application for analysing US equity performance across major historical financial crises

- Developed a FastAPI backend serving OHLC data for 3,158 US equity tickers (S&P 500, NASDAQ, NYSE, Forbes 2000) spanning 1970–2022
- Computed financial metrics per request including annualised volatility, average log return, ROI ratio, and recovery duration using pandas and NumPy
- Integrated Groq LLM to generate natural language summaries of computed metrics for a stock and time period
- Built an interactive Next.js frontend with Recharts visualisations for comparing stock performance across the Dot-Com crash, 2008 Financial Crisis, and COVID-19
- Aggregated yearly performance across 11 GICS sectors and implemented PDF export of charts and metrics
- Containerised the full stack with Docker Compose for consistent local and production deployment
- github.com/ahmadfahim2k/stock-analyzer-group-project 

Skills

Languages

Python, JavaScript, C, SQL

Backend

Node.js, Django, FastAPI

Tools

Git, Streamlit, Docker

Frontend

React, Tailwind CSS, Recoil, Storybook

ML / AI

PyTorch, Scikit-Learn, XGBoost, LangChain, ChromaDB, HuggingFace, Optuna, RAG